

VALJEVO, Serbia

WMO: 132690

Lat: 44.317N

Lon: 19.917E

Elev: 581

StdP: 14.39

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.2	18.2	8.8	8.8	14.7	13.4	11.1	19.9	12.7	38.7	11.1	39.4	1.9	250	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	93.5	70.3	90.3	69.8	87.1	68.9	73.6	86.5	71.9	84.7	70.5	82.9	5.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	111.8	79.5	68.0	105.1	77.1	66.5	99.7	75.6	37.5	86.1	36.0	84.9	34.7	82.7	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.6	9.6	7.8	5.2	98.0	6.9	3.9	0.3	100.8	-3.8	103.1	-7.7	105.3	-12.7	108.2		
			4.8	76.4	7.0	3.1	-0.2	78.6	-4.2	80.4	-8.1	82.1	-13.2	84.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	34.3	37.7	45.3	54.4	62.5	69.1	72.6	72.5	63.8	54.5	45.4	36.3	(6)
(7)		DBStd	15.45	8.62	9.22	8.32	6.98	6.37	6.41	5.49	5.88	6.69	7.68	8.31	8.23	(7)
(8)		HDD50	1727	487	350	192	38	2	0	0	0	44	183	431		(8)
(9)		HDD65	4732	951	763	611	322	124	32	7	9	102	333	588	890	(9)
(10)		CDD50	3234	2	6	46	170	389	574	699	699	414	183	46	6	(10)
(11)		CDD65	763	0	0	0	4	46	156	241	243	66	6	1	0	(11)
(12)		CDH74	7943	0	0	4	91	537	1655	2465	2461	638	90	2	0	(12)
(13)		CDH80	3102	0	0	0	15	113	618	1043	1102	204	7	0	0	(13)
(14)	Wind	WSAvg	3.1	2.5	3.1	3.7	3.7	3.4	3.3	3.2	2.9	2.6	2.6	2.6		(14)
(15)	Precipitation	PrecAvg	31.3	1.9	1.9	2.3	2.4	3.3	3.9	3.0	2.5	2.8	2.6	2.3	2.4	(15)
(16)		PrecMax	50.9	3.7	3.2	6.0	6.7	12.0	7.6	8.5	7.0	7.3	6.3	4.4	5.9	(16)
(17)		PrecMin	19.4	0.5	0.6	0.3	0.3	0.8	1.1	0.3	0.1	0.6	0.1	0.4	0.2	(17)
(18)		PrecStd	6.9	0.9	0.8	1.4	1.3	2.1	1.7	1.9	1.7	1.5	1.8	1.0	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.5	66.9	73.6	82.0	87.0	93.7	96.8	99.5	92.0	80.5	72.5	65.3	(19)
(20)		0.4%	MCWB	50.3	51.6	54.9	61.4	66.0	71.0	69.9	70.8	68.2	63.2	58.5	53.9	(20)
(21)		2%	DB	55.5	61.4	69.0	76.0	83.1	90.1	93.1	94.1	86.0	76.7	67.7	57.7	(21)
(22)		2%	MCWB	47.4	49.3	53.0	58.5	65.0	70.6	70.6	70.0	66.7	62.4	55.7	49.7	(22)
(23)		5%	DB	51.2	56.6	64.8	72.1	80.0	86.9	89.7	90.3	80.8	72.6	63.4	52.6	(23)
(24)		5%	MCWB	45.1	47.3	51.6	56.7	64.0	69.5	70.2	69.3	64.7	60.5	54.0	46.7	(24)
(25)		10%	DB	46.6	51.9	60.4	68.3	76.6	83.5	86.0	86.4	76.8	68.3	59.0	47.8	(25)
(26)		10%	MCWB	42.9	45.4	49.5	54.8	62.6	68.0	69.0	68.7	63.4	59.1	52.1	44.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.9	53.3	57.0	63.3	69.2	75.8	75.6	75.8	70.3	66.0	59.6	54.5	(27)
(28)		0.4%	MCDWB	58.9	63.4	70.1	77.7	81.3	87.8	89.2	88.0	85.5	77.1	69.4	63.5	(28)
(29)		2%	WB	48.4	50.2	54.2	60.0	66.8	72.9	73.2	73.5	68.4	63.6	56.9	50.1	(29)
(30)		2%	MCDWB	54.4	58.8	66.0	72.4	78.7	84.8	86.9	86.2	80.9	73.1	65.3	56.3	(30)
(31)		5%	WB	45.7	48.1	52.2	58.1	65.4	71.1	71.6	71.7	66.5	61.6	54.7	47.3	(31)
(32)		5%	MCDWB	50.3	55.4	63.0	68.9	76.9	82.8	84.6	84.6	77.3	70.1	61.8	52.0	(32)
(33)		10%	WB	43.1	45.8	50.2	56.3	63.8	69.4	70.1	70.1	64.8	59.6	52.5	44.5	(33)
(34)		10%	MCDWB	46.7	51.6	59.1	65.9	74.4	80.9	82.4	82.4	73.9	67.1	58.3	47.4	(34)
(35)	Mean Daily Temperature Range		MDBR	13.4	16.1	18.1	19.4	19.6	19.2	20.1	21.1	19.2	19.2	16.7	12.5	(35)
(36)		5% DB	MCDBR	22.6	26.8	27.9	28.1	26.7	26.3	27.8	29.7	27.6	26.8	24.7	21.7	(36)
(37)		5% DB	MCWBR	15.0	15.4	13.9	12.8	11.2	9.9	9.3	9.7	10.7	13.4	13.8	14.3	(37)
(38)		5% WB	MCDWR	21.0	23.9	24.6	23.7	23.7	23.1	23.6	24.0	23.6	23.8	22.2	20.0	(38)
(39)	5% WB	MCWBR	14.5	14.3	12.8	11.5	11.0	9.6	9.4	9.7	10.2	12.7	13.0	13.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.366	0.401	0.459	0.450	0.445	0.446	0.454	0.432	0.393	0.352	0.334	(40)	
(41)		taud	2.345	2.243	2.198	2.063	2.146	2.207	2.204	2.179	2.209	2.312	2.374	2.359	(41)	
(42)		Ebn at Noon	243	255	262	255	262	263	261	254	249	243	236	232	(42)	
(43)		Edh at Noon	26	34	40	50	47	45	44	44	39	31	25	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	415	666	1045	1395	1711	1906	1957	1746	1221	847	521	351	(44)	
(45)		RadStd	59	85	124	157	166	185	152	170	135	99	49	55	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.37	N/A	N/A	N/A	+1.13	+1.17	N/A	N/A	N/A	N/A	(46)
(47)	Regional (42 neighbors)	+1.19	N/A	N/A	N/A	+0.70	+0.81	N/A	-279	+196	+108	(47)

Nomenclature: See separate page